Option D – Energy and the Environment

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)	Use of $\lambda_{\max} = \frac{W}{T}$ (1) $\lambda_{\max} = \frac{2.9\times10^{-3}}{288} = 1\times10^{-5}$ [m] (1)	1	1		2	1	
		(ii)	Gases allow shorter λ solar radiation to pass [where it is absorbed by Earth] (1) Gases absorb [some of the] longer λ / infra-red / 10 μm radiation [emitted by Earth] (1) Gases emit radiation {in all directions / back to Earth} (1)	3			3		
	(b)	(i)	$\text{mass}_{\text{ice}} = 920\times3.0\times10^{11} = 2.76\times10^{14}$ **and** $V_{water} = \frac{2.76\times10^{14}}{1000} = 2.76\times10^{11}$ (1) $2.76\times10^{11}\times20 = 5.52\times10^{12}$ [m^3] (1) **Alternatively for 1st marking point:** $V_{\text{water}} = \frac{3.0\times10^{11}}{\frac{1000}{920}} = 2.76\times10^{11}$		2		2	2	
		(ii)	Darker areas will absorb more radiation / heat [than lighter areas] (1) Resulting in an increased temperature and rate of ice melting (1)		2		2		
	(c)	(i)	Hydroelectric relatively constant whereas PV increasing (1) No new hydroelectric sites available or commercially viable / advances in efficiency of PV cells / availability of [domestic] PV cells (1)			2	2		